

In-order

Post order

LNR

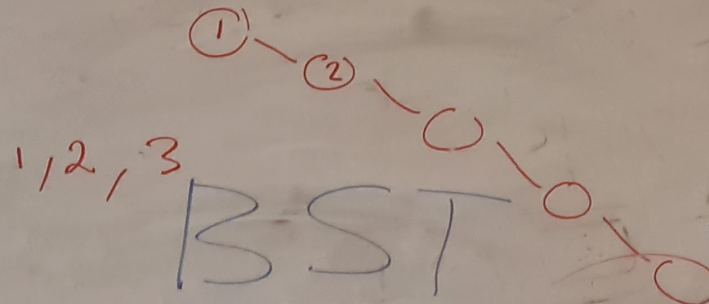
LRN

In-order

pre

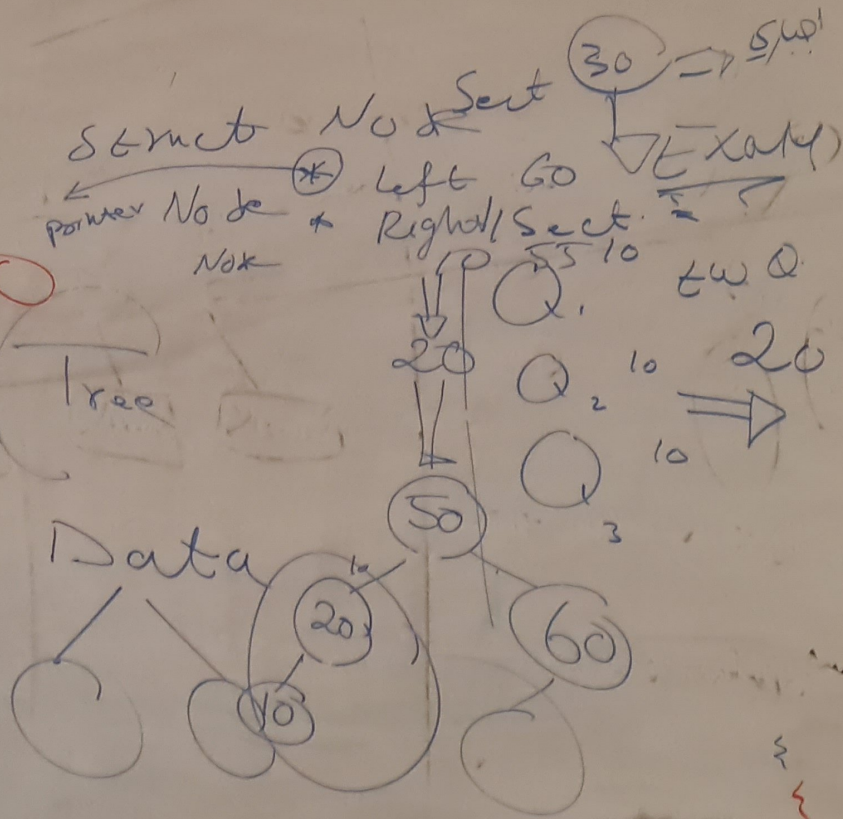
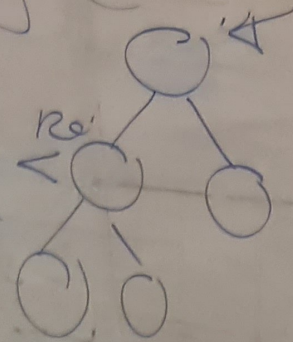
LNR

NLR



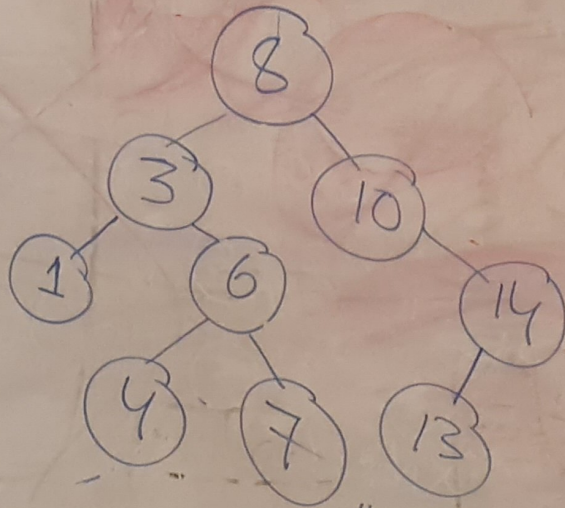
Binary Search Tree

- ① Root
- ② smallest left
- ③ Data right

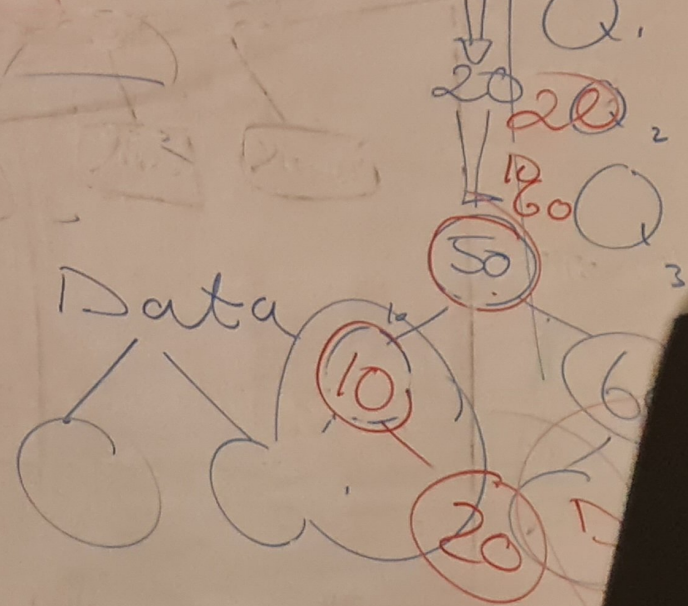


Construct BST from:

scan $\rightarrow$ 8, 3, 10, 1, 6, 14, 13, 4, 7

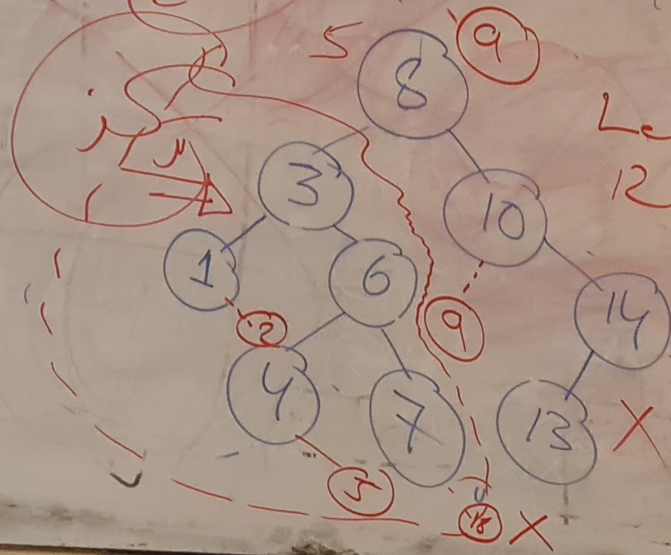


struct No & Sect
* Left Go
Pointer No de * Right / Sect
Nok



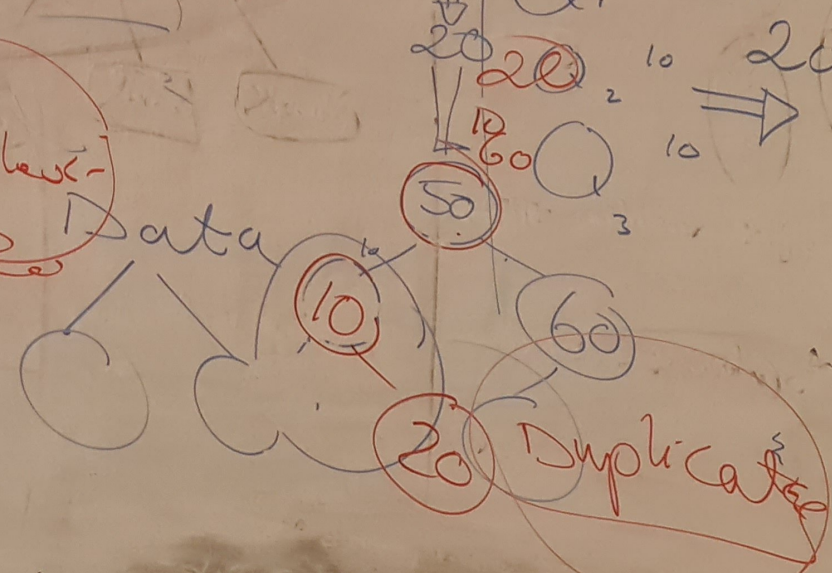
Construct BST from:

Scan $\rightarrow 8, 3, 10, 1, 6, 14, 13, 4, 7$



Red
Left $\rightarrow$ Smaller
Right $\rightarrow$ Bigger

Struct No & Sect $\rightarrow$ 30 $\Rightarrow$ 5/10
Left Go $\rightarrow$ Exact
Right Sect $\rightarrow$ 10
Pointer No de $\rightarrow$ 10
No x



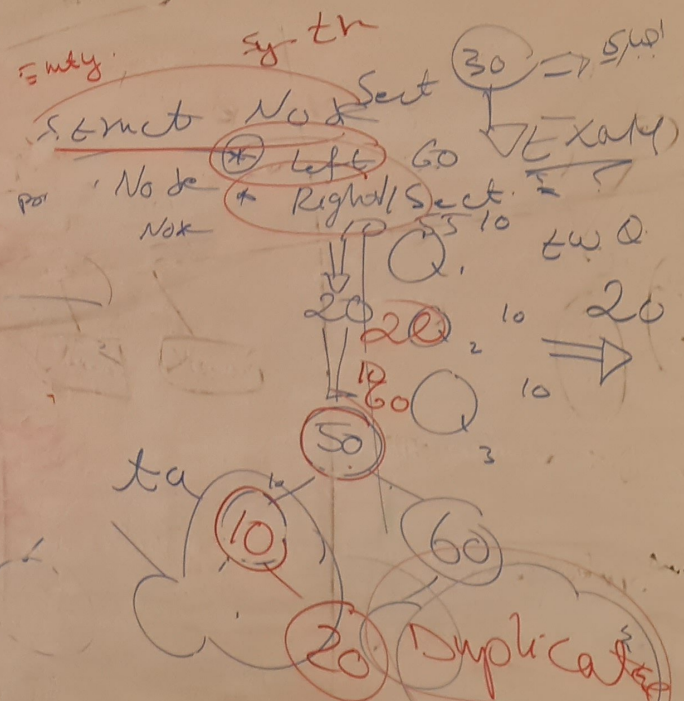
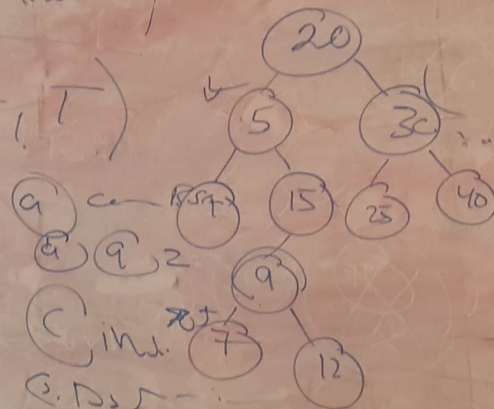
Duplicate

Search(X, T)

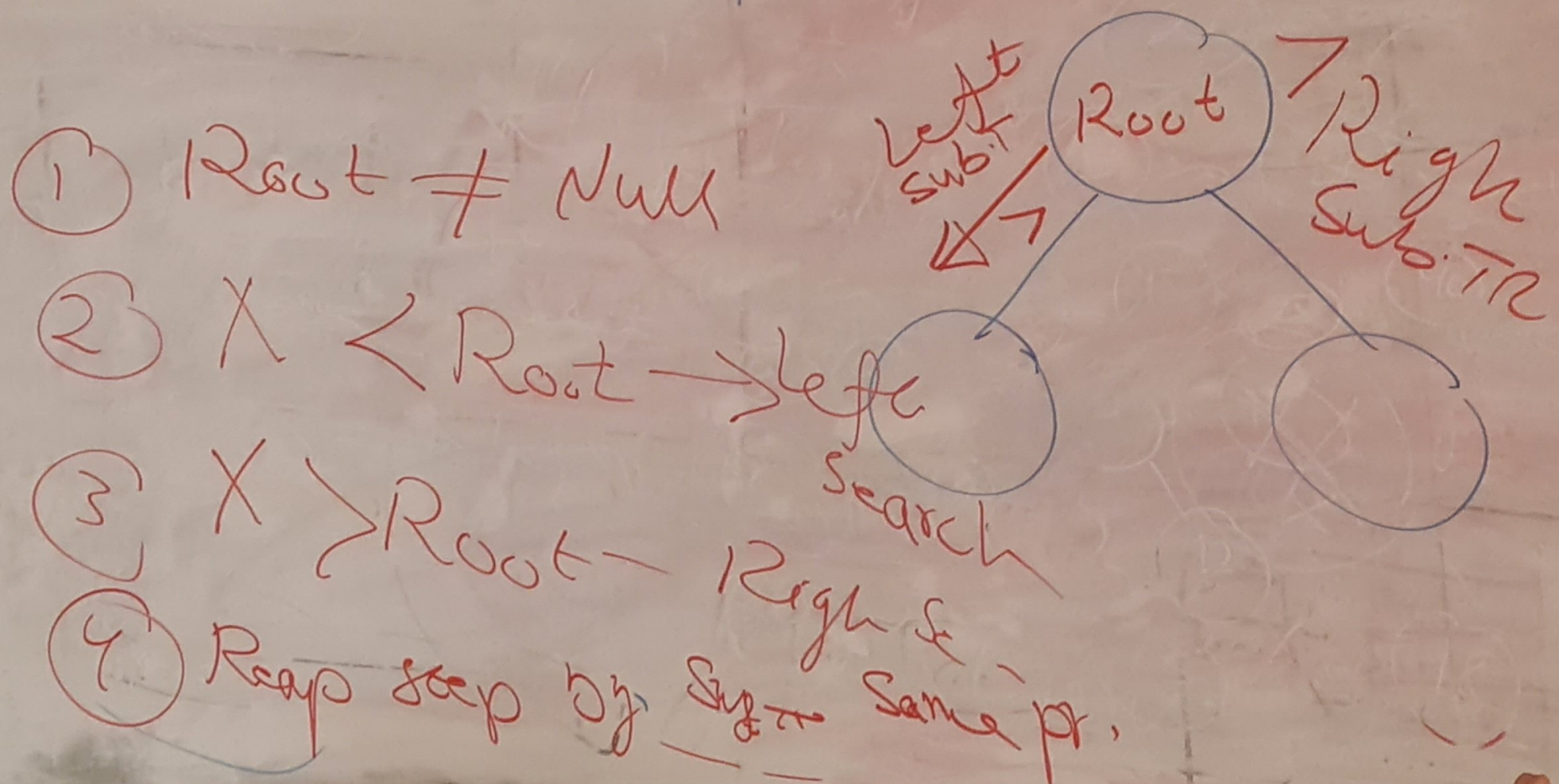
Root = ! null

* Search(7, T)

- ① $7 < 20 \rightarrow$ Left
- ② $7 > 5 \rightarrow$ Right
- ③ $7 < 15 \rightarrow$ Left
- $7 < 9 \rightarrow$ Left
- $7 = 7$ Found



Max, Min value BST



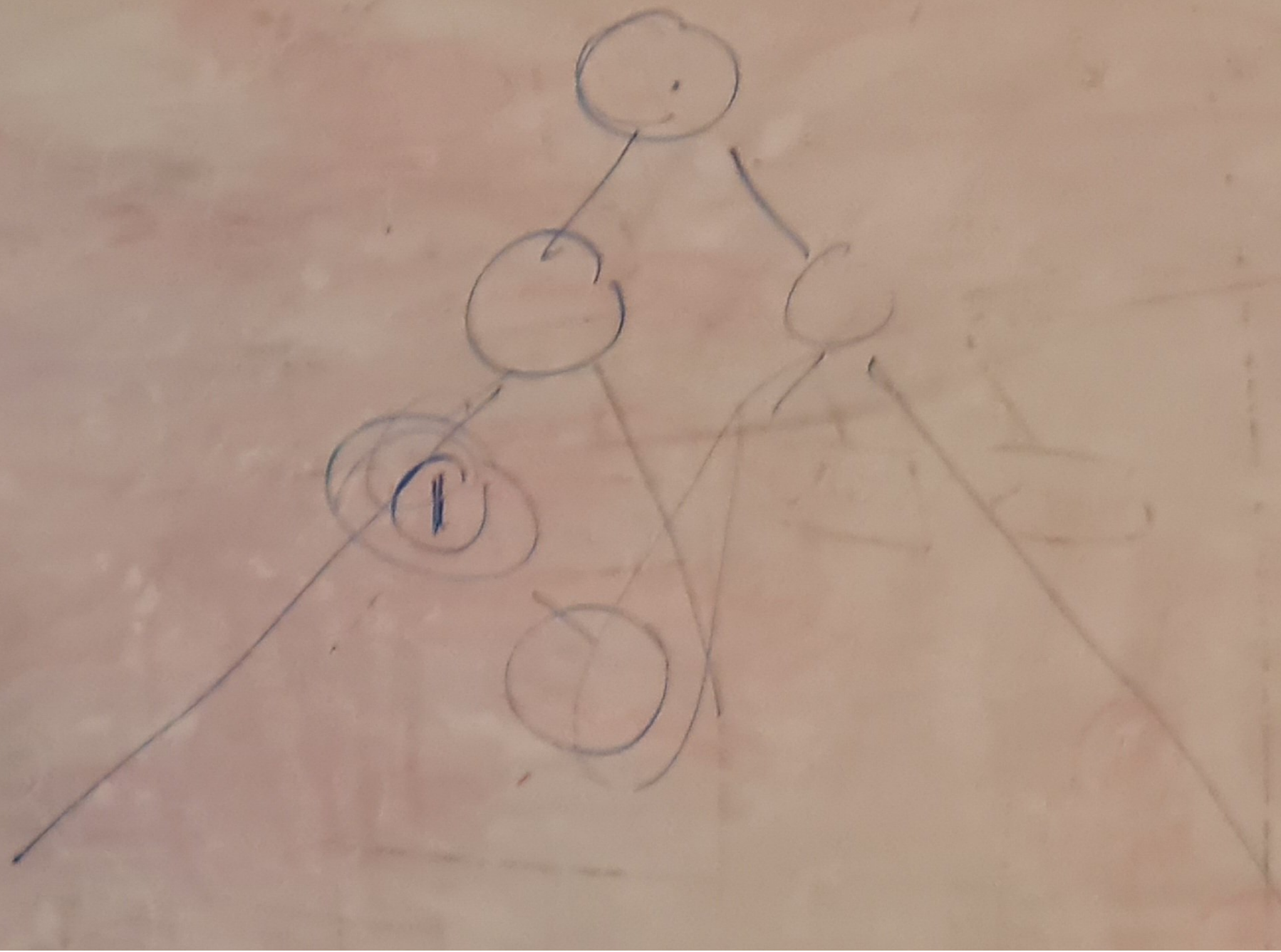
Max(n)

if (n.left == null)

return n

else

return Min(n.left)

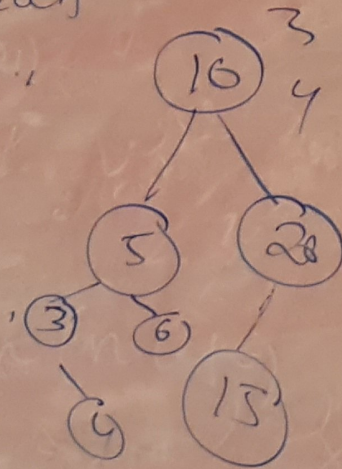


operation of BST

① insertion ^{Leaf}

15

② Deletion



6

Operation Of BST

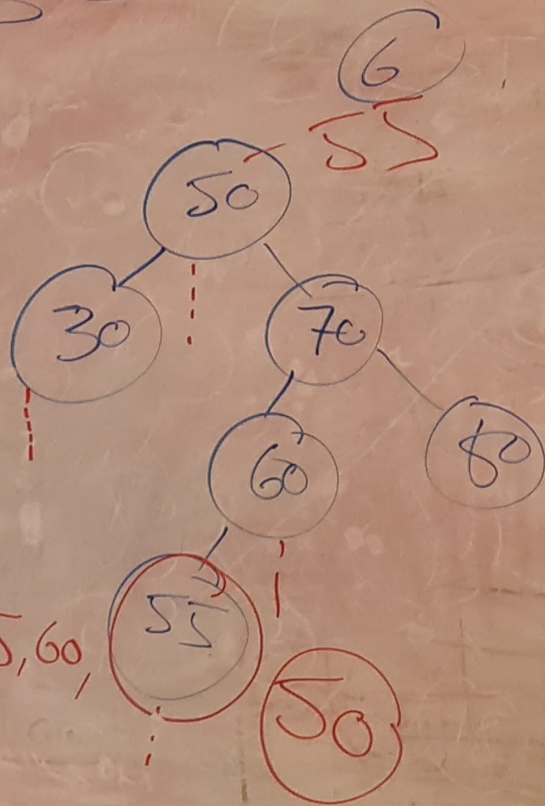
① insertion ^{Leaf}

15

② Deletion

Find
Inorder

30, 50, 55, 60,
70, 80



① Leaf
Delete - ② node with one child

③ node with two children

8

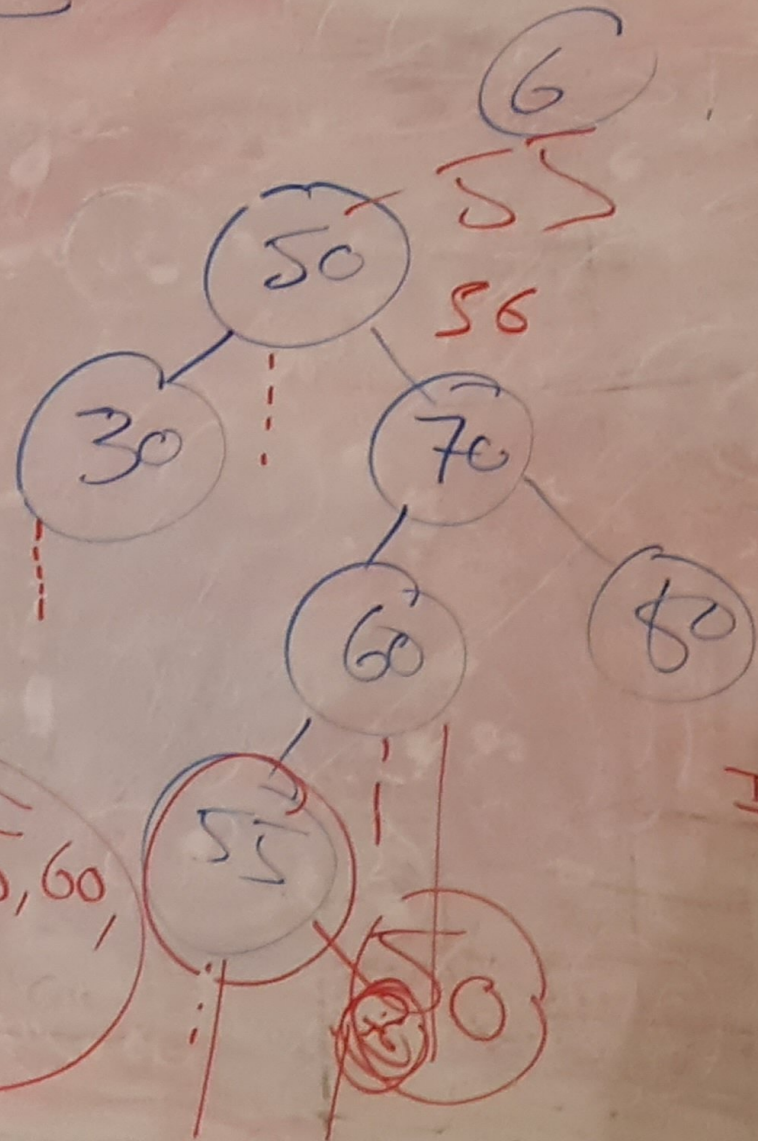
Operation Of BST

① insertion ^{Leaf}

15

② Deletion

Find Inorder $\leftarrow 30, 50, 55, 60, 70, 80$



(6)
55
56
70

* Inorder Successor!

Go to right then

leftmost (smallest value)